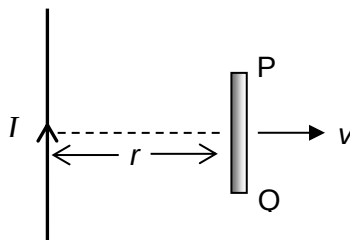


- 25 The diagram below shows a long wire carrying a current I .

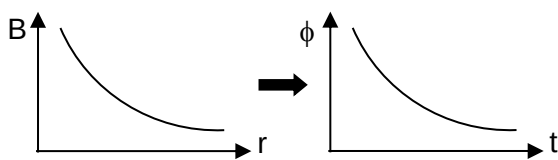
A straight conductor PQ is placed on the same vertical plane as the wire and is moved at constant speed v away from the wire.



How does the magnitude of the induced e.m.f. in PQ vary and which end is at a higher potential?

	magnitude of induced e.m.f.	end at higher potential
A	decreases	P
B	decreases	Q
C	increases	P
D	increases	Q

- 25 **A** Since rod is moving at constant v , graph of B vs r will be similar to graph of ϕ vs t



Since gradient drops as t increases, $\frac{d\phi}{dt}$ decreases as conductor PQ moves away
 $\rightarrow$ magnitude of induced emf decreases.
 Using FRHR, induced current is from Q to P, therefore P is at higher potential

26 The alternating current from an a.c. mains supply varies sinusoidally. The graph below